

CV of Minarul Islam

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Research Focus Area

- Natural Language Processing, Machine Learning, Deep Learning, Data Science, Reliable AI, and Cybersecurity-Oriented Text Analytics.
- Current research focuses on reliable and context-aware text classification, sentiment analysis, vulnerability severity classification, distribution-shift robustness, and knowledge-enriched decision support.
- Teaching interests include programming, data structures, software engineering, software quality assurance, data science, machine learning, natural language processing, and cybersecurity analytics.

Academic Preparation

- **Auburn University** Auburn, Alabama, USA
Ph.D. Student in Computer Science and Software Engineering, CGPA: 3.82/4.00
Aug 2022 – Present
Dissertation Committee Chair: Dr. Cheryl D. Seals
Dissertation Committee Members: Dr. Jakita Owenby Thomas, Dr. Richard Chapman, Dr. Daniela Marghitu, Dr. Shazibul Islam Shamim
Dissertation Focus: Reliable and Context-Aware Sentiment Analysis under Linguistic Complexity
- **Auburn University** Auburn, Alabama, USA
M.S. in Computer Science and Software Engineering, CGPA: 3.82/4.00
Aug 2022 – Summer 2026
- **Universiti Malaysia Pahang** Pahang, Malaysia
M.S. by Research in Electrical and Electronic Engineering
Sep 2018 – Feb 2021
Thesis: Non-Invasive Blood Glucose Estimation Using Ultra-Wideband Sensing and Machine Learning
- **Jessore University of Science & Technology** Jessore, Bangladesh
B.Sc. in Computer Science and Engineering
Feb 2012 – Nov 2016
Undergraduate Thesis: Ransomware File Encryption Attacks and Data Protection Framework

Peer-Reviewed Journal Articles

- **[IEEE Access 2025]**: Aish Albladi, **Minarul Islam**, and Cheryl Seals. “Sentiment Analysis of Twitter Data Using NLP Models: A Comprehensive Review.” *IEEE Access*, 2025.
- **[IEEE Access 2025]**: Aish Albladi, **Minarul Islam**, Amit Das, Maryam Bigonah, Zheng Zhang, Fatemeh Jamshidi, Mostafa Rahgouy, Nilanjana Raychawdhary, Daniela Marghitu, and Cheryl Seals. “Hate Speech Detection Using Large Language Models: A Comprehensive Review.” *IEEE Access*, 13:20871–20892, 2025.
- **[SN Applied Sciences 2020]**: Mamunur Rashid, **Minarul Islam**, Norizam Sulaiman, Bifta Sama Bari, and Md. Ripon. “Electrocorticography Based Motor Imagery Movements Classification Using Long Short-Term Memory Based on Deep Learning Approach.” *SN Applied Sciences*, 2:1–7, 2020.
- **[SN Applied Sciences 2020]**: **Minarul Islam**, Md Shawkat Ali, Nusrat Jahan Shoumy, Sabira Khatun, Mohamad Shaiful Abdul Karim, and Bifta Sama Bari. “Non-Invasive Blood Glucose Concentration Level Estimation Accuracy Using Ultra-Wide Band and Artificial Intelligence.” *SN Applied Sciences*, 2:1–9, 2020.

Peer-Reviewed Conference Articles

- **[IEEE International Workshop on Antenna Technology 2025]**: Mohammed Nazmus Shakib, Shaif Uddin Khondaker, Zebreail Hamden, Atiqur Rahman Uday, **Minarul Islam**, and Lim Heng Siong. “Design of a Compact Transparent Antenna for 5G Communication Systems.” In *2025 IEEE International Workshop on Antenna Technology*, pages 1–4. IEEE, 2025.
- **[ACM Conference on Embedded Networked Sensor Systems 2024]**: **Minarul Islam**, Xueyang Hu, and Tao Shu. “Poster: Vi-Detect: Fine-Grained Vibration-Based Component Looseness Detection Using Smartphones.” In *Proceedings of the 22nd ACM Conference on Embedded Networked Sensor Systems*, pages 822–823, 2024.
- **[IEEE International Conference on Sensing, Communication, and Networking 2024]**: Xueyang Hu, Tian Liu, **Minarul Islam**, and Tao Shu. “MMMC: A Contactless Wood Moisture Content Measurement System Based on COTS FMCW mmWave Radar.” In *2024 21st Annual IEEE International Conference on Sensing, Communication, and Networking*, pages 1–9. IEEE, 2024.
- **[IOP Conference Series: Materials Science and Engineering 2021]**: Md Zahangir Alam, Md Nazmus Sakib, **Minarul Islam**, and Mohammad Jalal Uddin. “Various Risks and Safety Analysis to Reduce Fire in Oil Refinery Plant.” In *IOP Conference Series: Materials Science and Engineering*, volume 1078, page 012028. IOP Publishing, 2021.
- **[IOP Conference Series: Materials Science and Engineering 2021]**: Md Nazmus Sakib, Md Mahmudul Hasan Sohel, Shahanul Islam, and **Minarul Islam**. “Smart Solution for Low Humidity Problems Using Automatic Ultrasonic Humidifier.” In *IOP Conference Series: Materials Science and Engineering*, volume 1078, page 012035. IOP Publishing, 2021.
- **[InECCE 2020]**: Bifta Sama Bari, Sabira Khatun, Kamarul Hawari Ghazali, Md Moslemuddin Fakir, Wan Nur Azhani W Samsudin, Mohd Falfazli Mat Jusof, Mamunur Rashid, **Minarul Islam**, and Mohd Zamri Ibrahim. “Ultra Wide Band Based Early Breast Cancer Detection Using Artificial Intelligence.” In *InECCE2019: Proceedings of the 5th International Conference on Electrical, Control and Computer Engineering*, pages 505–515. Springer, 2020.

- **[InECCE 2020]**: Bifta Sama Bari, Sabira Khatun, Kamarul Hawari Ghazali, Md Moslemuddin Fakir, Mohd Hisyam Mohd Ariff, Mohd Faizal Jamlos, Mamunur Rashid, **Minarul Islam**, Mohd Zamri Ibrahim, and Mohd Falfazli Mat Jusof. “Bandwidth and Gain Enhancement of a Modified Ultra-Wideband Micro-Strip Patch Antenna Using a Reflecting Layer.” In *InECCE2019: Proceedings of the 5th International Conference on Electrical, Control and Computer Engineering*, pages 463–473. Springer, 2020.
- **[Journal of Physics: Conference Series 2020]**: **Minarul Islam**, Chai Ly Min, Nusrat Jahan Shoumy, Md Shawkat Ali, Sabira Khatun, Mohamad Shaiful Abdul Karim, and Bifta Sama Bari. “Software Module Development for Non-Invasive Blood Glucose Measurement Using an Ultra-Wide Band and Machine Learning.” In *Journal of Physics: Conference Series*, volume 1529, page 052066. IOP Publishing, 2020.
- **[Cyber Security and Computer Science 2020]**: Bifta Sama Bari, Sabira Khatun, Kamarul Hawari Ghazali, **Minarul Islam**, Mamunur Rashid, Mostafijur Rahman, and Nurhafizah Abu Talip. “Performance Comparison of Early Breast Cancer Detection Precision Using AI and Ultra-Wideband Bio-Antennas.” In *Cyber Security and Computer Science: Second EAI International Conference, ICONCS 2020*, pages 354–365. Springer, 2020.
- **[Cyber Security and Computer Science 2020]**: **Minarul Islam**, Sabira Khatun, Nusrat Jahan Shoumy, Md Shawkat Ali, Mohamad Shaiful Abdul Karim, and Bifta Sama Bari. “Non-Invasive Diabetes Level Monitoring System Using Artificial Intelligence and UWB.” In *Cyber Security and Computer Science: Second EAI International Conference, ICONCS 2020*, pages 376–387. Springer, 2020.
- **[Emerging Technology in Computing, Communication and Electronics 2020]**: **Minarul Islam**, Sabira Khatun, Kamarul Hawari Ghazali, Mohd Mawardi Saari, Mohammed Nazmus Shakib, and Mohamad Shaiful Abdul Karim. “Performance Comparison of Non-Invasive Blood Glucose Level Using Artificial Neural Network and Ultra-Wide Band Antenna.” In *2020 Emerging Technology in Computing, Communication and Electronics*, pages 1–5. IEEE, 2020.
- **[InECCE 2020]**: NJ Shoumy, DM Rahaman, S Khatun, WN Azhani, MH Ariff, MN Morshed, **Minarul Islam**, SNA Manap, and MFM Jusof. “Multi-Hop File Transfer in WiFi Direct Based Cognitive Radio Network for Cloud Backup.” In *InECCE2019: Proceedings of the 5th International Conference on Electrical, Control and Computer Engineering*, pages 389–400. Springer, 2020.

Manuscripts Under Review and In Preparation

- **[ACM Transactions on Sensor Networks]**: **Minarul Islam**, Xueyang Hu, and Tao Shu. “Vi-Detect: Fine-Grained Vibration-Based Component Looseness Detection Using Smartphones.” Minor correction submitted.
- **[PeerJ Computer Science]**: Aish Albladi, M. K. Uddin, **Minarul Islam**, and Cheryl Seals. “TWSSenti: A Novel Hybrid Framework for Topic-Wise Sentiment Analysis on Social Media Using Transformer Models.” Submission in progress.
- **[Journal of Information Security and Applications]**: **Minarul Islam** and Tao Shu. “DeepFakeDetect: Fine-Grained Deepfake Audio Detection Using Mechanical Modeling of Vocal Folds.” Under review.
- **[IEEE Transactions on Intelligent Transportation Systems]**: **Minarul Islam**, Aish Albladi, Mohammad Arafatur Rahman, Muhammad Asadur Rahman, and Cheryl Seals. “Reconstruction Accuracy and Forecasting Utility in Traffic Data Imputation: A Downstream Evaluation Study.” In preparation.
- **[IEEE International Conference on Machine Learning and Applications 2026]**: **Minarul Islam**, Aish Albladi, Muhammad Asadur Rahman, and Cheryl Seals. “ShiftAware-Select: Reliable NLP Classification under Distribution Shift with Abstain-to-Human Decisioning.” Under review.
- **[IEEE International Conference on Machine Learning and Applications 2026]**: **Minarul Islam**, Aish Albladi, Muhammad Asadur Rahman, and Cheryl Seals. “CAGLSAv2: Position-Invariant Discourse-Span Attention for Length-Robust Sentiment Analysis.” Under review.
- **[IEEE/ACM Transactions on Audio, Speech, and Language Processing]**: **Minarul Islam**, Aish Albladi, Muhammad Asadur Rahman, and Cheryl Seals. “LayerLens: Task-Relevant Layer Disruption Analysis for Self-Supervised Speech Encoders.” Submission in progress.
- **[IEEE AIBThings 2026]**: **Minarul Islam**, Aish Albladi, Muhammad Asadur Rahman, and Cheryl Seals. “Irony-Aware Twitter Sentiment Analysis via Multi-Task Learning.” Submission in progress, IEEE AIBThings 2026: International Conference on Artificial Intelligence, Blockchain, and Internet of Things.
- **[IEEE AIBThings 2026]**: **Minarul Islam**, Aish Albladi, Muhammad Asadur Rahman, and Cheryl Seals. “Hierarchical Severity Classification of Vulnerability Disclosures with Automatic Common Weakness Enumeration Enrichment.” Submission in progress, IEEE AIBThings 2026: International Conference on Artificial Intelligence, Blockchain, and Internet of Things.
- **[IEEE AIBThings 2026]**: **Minarul Islam**, Aish Albladi, Muhammad Asadur Rahman, and Cheryl Seals. “DualGraph-RoBERTa: Fusing Syntactic and Semantic Graphs for Aspect-Based Sentiment Analysis.” Submission in progress, IEEE AIBThings 2026: International Conference on Artificial Intelligence, Blockchain, and Internet of Things.

Notable Research Projects

- **ShiftAware-Select: Reliable NLP Classification under Distribution Shift with Abstain-to-Human Decisioning**
Text classifiers often lose reliability when deployed on data from a different platform, domain, or writing style. Overconfident errors are especially important in moderation, safety, and human-review settings. We developed a post-hoc reliability layer using calibrated confidence and entropy-based uncertainty to decide whether a prediction should be accepted automatically or routed to human review. On noisy and shifted toxicity datasets, the method reduced overconfident errors by up to **13.7%**, improved selective risk by up to **49.9%**, and maintained a stable **65%** automatic acceptance rate.
- **CAGLSAv2: Position-Invariant Discourse-Span Attention for Length-Robust Sentiment Analysis**

Sentiment models often struggle when opinions depend on contrastive clauses, such as cases involving “but” or “however.” These cases matter because the final opinion may depend more on clause structure than on individual sentiment words. We designed a discourse-aware transformer model that represents the full sentence, the clause before the contrast marker, and the clause after the contrast marker. The model achieved **0.8661**, **0.9621**, and **0.9071** Macro-F1 on TweetEval, SST-2, and MR.

- **Irony-Aware Twitter Sentiment Analysis via Multi-Task Learning**

Social media sentiment is difficult because users often express opinions indirectly through irony, sarcasm, or emotion. Literal word matching can misread such posts and produce unreliable sentiment labels. We developed a multi-task transformer framework that jointly learns sentiment, irony, and emotion using a shared Twitter-RoBERTa encoder. The model achieved **0.7241 Macro-F1** on TweetEval Sentiment and outperformed BERT-base, RoBERTa-base, DeBERTa-base, and single-task Twitter-RoBERTa baselines. An optional irony-conditioned module added only **0.474%** parameters and reached **0.8699 AUC-ROC**.

- **Hierarchical Severity Classification of Vulnerability Disclosures with Automatic Common Weakness Enumeration Enrichment**

Software vulnerability reports are often short, technical, and difficult to prioritize. Delayed prioritization can slow responses to high-risk vulnerabilities. We developed a knowledge-enriched classification framework that adds Common Weakness Enumeration context to vulnerability descriptions before severity prediction. The study evaluated **158,379** National Vulnerability Database records, enriched **84.3%** of them with CWE information, and achieved **0.7658 Macro-F1** across low, medium, high, and critical severity classes.

- **DualGraph-RoBERTa: Fusing Syntactic and Semantic Graphs for Aspect-Based Sentiment Analysis**

Aspect-based sentiment analysis requires a model to identify opinion toward a specific target, not only the overall sentiment of a sentence. A review may praise one aspect while criticizing another. We developed a dual-graph framework that combines syntactic dependency relations with semantic co-occurrence relations to better connect each aspect with its opinion evidence. The model achieved **0.8994**, **0.8763**, and **0.8403** Macro-F1 on SemEval-2014 Laptop, SemEval-2014 Restaurant, and SemEval-2016 benchmarks.

- **M.S. Thesis: Non-Invasive Blood Glucose Estimation Using Ultra-Wideband Sensing and Machine Learning**

My M.S. thesis addressed frequent blood glucose monitoring, where standard finger-prick methods require invasive blood sampling and can be uncomfortable for repeated use. I developed a machine learning-based framework using ultra-wideband sensing data to estimate blood glucose concentration without direct blood sampling. The work included data preprocessing, feature preparation, model training, model comparison, and performance evaluation. The outcome was an applied biomedical AI framework that connected sensing data with non-invasive glucose-level prediction.

- **Undergraduate Thesis: Ransomware File Encryption Attacks and Data Protection Framework**

My undergraduate thesis addressed ransomware attacks, where malicious software encrypts files and blocks access to important data. The main problem was the lack of secure backups, recovery procedures, and early protection practices among many users and small organizations. I analyzed infection paths, file-encryption behavior, user-level security weaknesses, recovery limitations, and prevention strategies. The outcome was a practical data protection framework organized around prevention before infection, damage reduction during an attack, and recovery after file encryption.

Research and Professional Experience

- **Auburn University**

Auburn, Alabama, USA

Graduate Research Assistant, Department of Computer Science and Software Engineering

Aug 2022 – Present

- Conduct research on reliable NLP, sentiment analysis, toxicity detection, discourse-aware modeling, multi-task learning, graph-based NLP, knowledge-augmented classification, and LLM-style evaluation.
- Design and evaluate transformer-based models using public benchmark datasets, ablation studies, calibration analysis, robustness testing, and distribution-shift evaluation.
- Prepare research manuscripts, experimental protocols, reproducibility materials, technical presentations, and response documents for peer-reviewed journal and conference submissions.
- Collaborate with faculty and graduate researchers on dataset preparation, method design, result analysis, manuscript revision, and research documentation.

- **Universiti Malaysia Pahang**

Pahang, Malaysia

Research Officer; project assigned to Sime Darby Plantation

Aug 2020 – Jan 2021

- Worked on applied AI research projects in collaboration with Sime Darby Plantation, with emphasis on computer vision and deep learning.
- Contributed to palm fruit bunch object detection from video using deep transfer learning, including YOLOv4-tiny and Darknet-based detection workflows.
- Supported deep learning model development for land surface and slope deformation analysis through data preparation, model testing, and result interpretation.
- Visited Selangor plantation sites to understand field conditions, data collection constraints, and deployment requirements for agricultural AI applications.

- **Universiti Malaysia Pahang**

Pahang, Malaysia

Graduate Research Assistant

Sep 2018 – Feb 2021

- Developed machine learning and deep learning models for biomedical sensing, classification, signal-based prediction, ultra-wideband applications, and applied AI research.
- Built a machine learning framework for non-invasive blood glucose estimation using ultra-wideband sensing data.

- Collaborated on early breast cancer detection research using ultra-wideband sensing and artificial intelligence.
- Worked on grape leaf disease detection using deep learning and image-based classification.

Teaching Experience

- **Auburn University** Auburn, Alabama, USA
Graduate Teaching Assistant, Department of Computer Science and Software Engineering Aug 2022 – Present
Courses Supported: Senior Design Project; Software Quality Assurance.
 - Supported Senior Design Project across Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, and Spring 2026.
 - Supported a large senior capstone course serving more than 100 undergraduate students per semester in multiple concurrent project teams.
 - Mentored more than a dozen sponsor-driven student team projects each semester across databases, machine learning, deep learning, mobile applications, web applications, front-end interfaces, back-end services, and full-stack systems.
 - Helped student teams understand sponsor and customer requirements, translate non-technical needs into clear system features, and deliver outcomes aligned with stakeholder expectations.
 - Supported Software Quality Assurance for undergraduate and graduate students through grading, project feedback, test design instruction, and office-hour support.
- **Universiti Malaysia Pahang** Pahang, Malaysia
Teaching Assistant and Laboratory Support Sep 2018 – Feb 2021
Courses Supported: Digital Signal Processing; Senior Design Project; Database Management; Deep Learning.
 - Assisted laboratory activities, grading, project mentoring, and office-hour support.
 - Supported students in MATLAB and Python implementation of signal processing and deep learning algorithms.
- **Model Polytechnique Institute** Jessore, Bangladesh
Lecturer Jan 2017 – Jul 2018
Courses Taught: Basic Electrical Engineering I; Programming in C; Data Structures.
 - Delivered lectures, laboratory sessions, assignments, quizzes, and examinations for foundational computing and engineering courses.
 - Guided students in programming fundamentals, structured problem solving, arrays, pointers, linked lists, stacks, queues, and sorting methods.

Awards, Scholarships, and Achievements

- **Master Research Scheme Scholarship**, Universiti Malaysia Pahang. Awarded to support graduate research.
- **Postgraduate Research Scheme Scholarship**, Grant No. PGRS1903146, Universiti Malaysia Pahang. Awarded to support research and conference participation.
- **Best Innovative Engineering Technology Award**, Creation, Innovation, Technology & Research Exposition, Universiti Malaysia Pahang, 2020.
- **Gold Medal**, Creation, Innovation, Technology & Research Exposition, Universiti Malaysia Pahang, 2020.
- **Silver Medal**, Creation, Innovation, Technology & Research Exposition, Universiti Malaysia Pahang, 2020.
- **Bronze Medal**, Creation, Innovation, Technology & Research Exposition, Universiti Malaysia Pahang, 2019.

Professional Service and Outreach Activities

- Reviewer: IEEE International Conference on Machine Learning and Applications.
- Reviewer: *IEEE Access*.
- Reviewer: *IEEE International Conference on Computer Communications*.
- Reviewer: *Journal of Applied Artificial Intelligence*.
- Reviewer: *Cloud Computing* published by Springer.
- Presenter and committee member: 8th IEEE Symposium on Acoustics, Speech and Signal Processing, Universiti Malaysia Pahang, 2020.
- Presented research during the Alabama State University visit to Auburn University, contributing to academic outreach and research exchange.
- Participant and presenter: Graduate Engineering Research Showcase Poster Presentation, Auburn University.
- Presented applied AI and engineering research at Creation, Innovation, Technology & Research Exposition, Universiti Malaysia Pahang, 2019–2020.
- Committee member: Applied Electronics and Computer Engineering Research Group, Universiti Malaysia Pahang.
- Volunteer: FOX HUNTING, Faculty of Electrical and Electronics Engineering Technology, Universiti Malaysia Pahang, 2018.
- Participant: Colloquium 2019, Universiti Malaysia Pahang.
- Participant: Auburn Hackathon, 2023.

Technical and Educational Tools

- **Programming and Data Tools:** Python, C, C++, Java, MATLAB, SQL, NumPy, Pandas, Scikit-learn, Matplotlib, Jupyter Notebook.
- **Machine Learning Tools:** PyTorch, TensorFlow/Keras, Hugging Face Transformers, model fine-tuning, text classification, model evaluation, and reproducible experiment workflows.
- **Research Writing Tools:** LaTeX, Overleaf, GitHub, Microsoft Office, and Google Workspace.
- **Educational Platforms:** Canvas, Blackboard, Zoom, Microsoft Teams, Google Classroom, and Turnitin.

References

Dr. Cheryl D. Seals

Professor

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Dr. Jakita O. Thomas

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Dr. Shazibul Islam Shamim

Tenure Track Assistant Professor

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Game Design and Development
Kennesaw State University
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